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Short communication

Novel $Zr_{1-3x}Y_xGd_xNb_xO_2$ medium-entropy oxide ceramics for advanced thermal barrier coatings

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ABSTRACT

A novel series of medium-entropy oxide ceramics, $Zr_{1-3x}Y_xGd_xNb_xO_2$ (x = 0.1~0.25), has been designed for advanced thermal barrier coatings (TBCs) applications. Through comprehensive characterization, the materials' thermal conductivity, phase composition, fracture toughness, and calcium-magnesium-alumino-silicate (CMAS) corrosion resistance were systematically investigated. The results revealed that low-doped compositions (x = 0.1, 0.125) maintained a stable single cubic phase structure, demonstrating remarkably low thermal conductivity ranging from 1.07 to 1.37 W/m·K at 1673 K. Notably, these materials exhibited exceptional phase stability, with no phase transformation observed even after prolonged exposure (300 h) at 1873 K. While the elastic modulus showed a gradual increase with extended heat treatment (remaining within 10~20 GPa after 300 h), the ceramics achieved a remarkable fracture toughness of 5.49 $MPa{\cdot}m^{1/2}$ coupled with outstanding CMAS corrosion resistance. Through strategic rare-earth element compositional design, a simultaneous enhancement of thermal, structural integrity, and mechanical properties was achieved, presenting a promising candidate material for next-generation aero-engine TBCs.

1. Introduction

The advancement of next-generation aero-engines demands operation under extreme conditions, requiring thrust-to-weight ratios exceeding 12~13 while enduring turbine inlet temperatures approaching 2000 K. These severe thermal-mechanical conditions expose the inherent limitations of conventional nickel-based superalloys. Yttria-stabilized zirconia (YSZ) thermal barrier coatings (TBCs) play a critical role by lowering substrate temperatures and enhancing thermal efficiency. However, existing YSZ coatings cannot sustain prolonged service above 1200 °C due to multiple challenges: phase instability [1,2], high thermal conductivity, poor sintering resistance, and susceptibility to CMAS corrosion [3,4]. Therefore, developing next-generation TBCs materials with superior performance has become a paramount research priority.

In recent years, multicomponent oxide ceramic systems have gained significant attention as promising candidates for advanced TBCs applications [5–7]. For instance, Chen et al. [8] reported a new fluorite-type high-entropy oxide $(Ce_{0.2}Zr_{0.2}Hf_{0.2}Sn_{0.2}Ti_{0.2})O_2$ with a room-temperature thermal conductivity of 1.28 W/m·K, half that of 7YSZ. Yao et al. [9] developed $Zr_{1-4x}Y_xYb_xTa_xNb_xO_2$, which exhibited fracture toughness as high as 4.59 $MPa{\cdot}m^{1/2}$. However, most of newly developed TBCs materials, while exhibiting excellent specific performance, generally demonstrate inferior comprehensive properties compared to YSZ. Moreover, the significant rare-earth element loading results in prohibitive costs, limiting practical implementation.

To overcome the above limitations, a novel series of $Zr_{1-3x}Y_xGd_xNb_xO_2$ medium-entropy ceramics was designed. The strategic selection of Gd^{3+} and Nb^{5+} as co-stabilizers leverages their extreme ionic radius disparity to synergistically enhance TBCs performance [10]. The large Gd^{3+} ions induce compressive lattice strain while the small Nb^{5+} ions create tensile distortion, collectively maximizing phonon

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scattering for ultralow thermal conductivity. Simultaneously, Nb^{5+} substitution generates oxygen vacancies that stabilize the cubic phase during prolonged 1873 K exposure, while Gd^{3+} enables dense apatite layer formation for exceptional CMAS resistance [11,12]. This dual-ion approach further strengthens grain boundaries, elevating fracture toughness beyond conventional TBCs materials. This multicomponent system is engineered to simultaneously achieve exceptional high-temperature phase stability, ultralow thermal conductivity, superior CMAS corrosion resistance, and enhanced mechanical properties, facilitating TBCs application in next-generation aero-engines.

2. Materials and experimental methods

Four multi-component $Zr_{1-3x}Y_xGd_xNb_xO_2$ oxide ceramics were designed, and the composition ratio scheme is shown in Table 1. The samples were fabricated by solid-phase reaction method, in which the original powders (China Metallurgical Research Institute, Beijing, China) of zirconia (99.99 %), yttria (99.99 %), gadolinia (99.99 %), and niobiumia (99.999 %) were weighed, mixed with ethanol as a grinding medium, and ground in a planetary ball mill for 20 h. The mixed and dried powders were pressed and moulded at a pressure of 500 MPa with a holding time of 10 min. The resulting samples were placed in a chamber muffle furnace and heated up to 1873 K at 6 K/min, and sintered in air for 20 h and then slowly cooled with the furnace.

The phase compositions of samples were characterized using X-ray diffractometry (XRD; D8 ADVANCE ECO, Bruker Technologies GmbH, Germany) using Cu-Kα ($\lambda = 1.5418°$) radiation within $10° \leq 2\theta \leq 90°$ with a scanning step size of 0.008°, and the phase composition was determined by image analysis software including JADE 6.0 and GSAS II, which were used to calibrate the phase content and cell parameters. The microstructures and morphologies of the samples were characterized by scanning electron microscopy (SEM; SU3500, Hitachi, Japan). Elemental analysis was performed using energy-dispersive X-ray spectroscopy (EDS; Ultim Max100, Oxford, UK) attached to the SEM. The densities and porosities of the samples were measured by Archimedes drainage method and image analysis, and the average value of porosity was calculated by counting 10 cross-sectional SEM images with a magnification of × 500 using Image-Pro®Plus v 6.0 [13,14] software.

Vickers microhardness measurements were performed on the polished surfaces (Ra ≤0.8 μm) using a 10 N load (HRMS-45 tester, Chongqing Lieb Instrument Co., Ltd.). Fracture toughness were calculated from the Vickers indentation pattern [15]. The elastic modulus and hardness were determined by nanoindentation (Nanoindentation, Hysitron Ti950, Bruker Technologies GmbH, Germany). Tests were conducted on polished surfaces with controlled parameters: 10 nm/s loading/unloading rate, 10 s hold time, and ≥10 indentations per sample to ensure statistical reliability.

The thermal diffusion coefficient (α) of $Zr_{1-3x}Y_xGd_xNb_xO_2$ was measured in the range of 673 K~1673 K at intervals of 473 K using a laser method thermal conductivity analyzer (LFA 427, Netzsch, Bavaria, Germany). The front and back surfaces of the samples were carbon-sprayed before and after the tests to absorb the heat of the laser pulse. Three measurements were taken for each sample at selected temperatures. The density (ρ) was determined by the Archimedes drainage method. The specific heat capacity (C_p) was calculated from the heat capacity values of the constituents according to the Neumann-Kopp rule [16,17], which has the core principle to treat the specific heat capacity of the two-phase material as the mass-fraction-weighted sum of the specific heat capacities of its constituent phases (e.g., cubic phase c-ZrO_2 and monoclinic phase m-$YNbO_4$):

$$C_p = f_{\text{cub}} \times C_{p,\text{cub}} + f_{\text{mon}} \times C_{p,\text{mon}}$$

In this formula, f_{cub} and f_{mon} represent the mass fractions of the cubic phase (c-ZrO_2) and monoclinic phase (m-$YNbO_4$), while $C_{p,cub}$ and $C_{p,mon}$ represent the specific heat capacities of the cubic phase (c-ZrO_2) and monoclinic phase (m-$YNbO_4$), respectively. The thermal performance data of all samples were shown in Table S1–S4. The thermal conductivity (λ) was calculated as the product of α, C_p and ρ, expressed by the equation: $\lambda = \alpha \times C_p \times \rho$ [17].

The phase stability of $Zr_{1-3x}Y_xGd_xNb_xO_2$ oxides was assessed under long-term thermal exposure at high temperatures by comparing XRD patterns before and after heat treatment at 1873 K for 100~300 h. During CMAS corrosion testing, the CMAS powder was weighed and uniformly coated on the polished surface of each sample, with a loading concentration close to 30 mg/cm^2 [18]. The samples with CMAS powder were placed in a chamber muffle furnace and heat-treated at 1673 K for different times to investigate the CMAS melt infiltration.

3. Results and discussion

3.1. Microstructure and phase composition

The phase compositions of $Zr_{1-3x}Y_xGd_xNb_xO_2$ ceramics were determined by Rietveld refinement of XRD patterns, as shown in Fig. 1a~d. The refinement yielded good agreement between the experimental and calculated patterns, with reliability factors of $R_{wp} < 13$ % and $R_p < 10.5$ %, confirming the accuracy of the structural analysis. The equimolar fraction $Zr_{0.25}Y_{0.25}Gd_{0.25}Nb_{0.25}O_2$ (x = 0.25, denoted as c-1) consisted of 81.5 % cubic phase and 18.5 % monoclinic phase. The formation of the cubic phase (c-ZrO_2) originates from the incorporation of the rare-earth ions (Y^{3+}, Gd^{3+}, Nb^{5+}) via partial substitution of Zr^{4+}, and the rare earth ions preferentially occupied the ZrO_2 lattice interstitials or substitutional sites to stabilize the cubic phase through oxygen vacancies. Since the rare-earth doping exceeded the solid solubility limit, the excess rare-earth elements could not completely enter the lattice. In particular, due to its smaller ionic radius and higher charge, Nb^{5+} had a slower diffusion rate than the other rare-earth ions, tended to accumulate at the grain boundaries, and promoted the formation of the monoclinic phase (m-$YNbO_4$).

The difference in diffusion rates of rare earth elements resulted in the accumulation of lattice stresses and drove the compositional segregation, further exacerbating the coexistence of the cubic and monoclinic phase. The non-equimolar component $Zr_{0.499}Y_{0.167}Gd_{0.167}Nb_{0.167}O_2$ (x = 0.167, denoted as c-2) was composed of 93.3 % cubic phase and 6.7 % monoclinic phase. Compared with sample c-1, the reduced total rare-earth doping concentration alleviated the solid-solution limit and mitigated the competitive diffusion effect among elements. This promoted a more uniform distribution of rare-earth ions within the ZrO_2 lattice, suppressed Nb segregation at grain boundaries, and consequently weakened the formation of m-$YNbO_4$. As a result, the cubic phase fraction significantly increased. For samples c-3 (x = 0.125) and c-4 (x = 0.1) with further reduced rare-earth doping content, the XRD refinement results indicated that their phase compositions were solely single cubic phases (c-ZrO_2, with >99 %). Since the total rare earth doping was

Table 1
The designed composition of all $Zr_{1-3x}Y_xGd_xNb_xO_2$ oxides, along with densities, porosities, grain sizes, Vicker's hardness, Young's modulus.

Sample	Composition	Density (g/cm^3)	Porosity (%)	Grain size (μm)	Hardness (GPa)	Young's modulus (GPa)
c-1	$Zr_{0.25}Y_{0.25}Gd_{0.25}Nb_{0.25}O_2$	6.4 ± 0.2	2.2 ± 0.5	7.9 ± 2	16 ± 3	109 ± 6
c-2	$Zr_{0.499}Y_{0.167}Gd_{0.167}Nb_{0.167}O_2$	6.3 ± 0.3	1.5 ± 0.4	13 ± 5	16 ± 5	150 ± 8
c-3	$Zr_{0.625}Y_{0.125}Gd_{0.125}Nb_{0.125}O_2$	6.1 ± 0.1	1.1 ± 0.2	19 ± 6	16 ± 1	192 ± 3
c-4	$Zr_{0.7}Y_{0.1}Gd_{0.1}Nb_{0.1}O_2$	6.0 ± 0.1	1.2 ± 0.2	22 ± 8	17 ± 1	200 ± 6

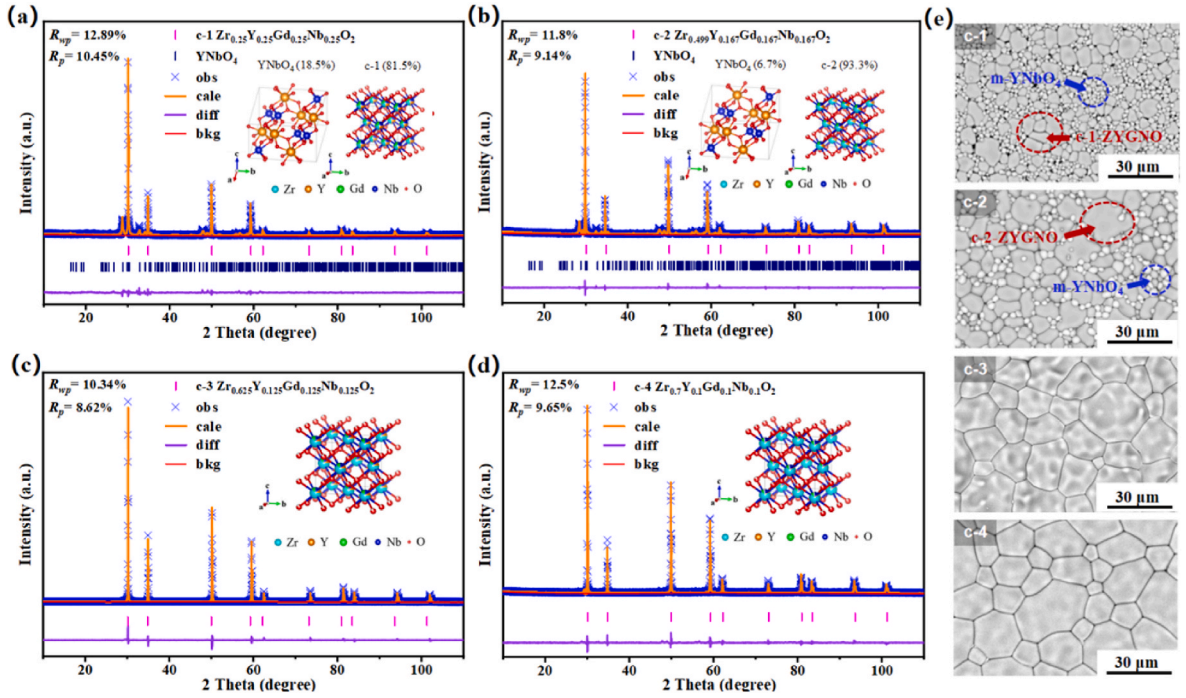


Fig. 1. Phase structure: Rietveld refinement of XRD patterns of (a) c-1 $Zr_{0.25}Y_{0.25}Gd_{0.25}Nb_{0.25}O_2$, (b) c-2 $Zr_{0.499}Y_{0.167}Gd_{0.167}Nb_{0.167}O_2$, (c) c-3 $Zr_{0.625}Y_{0.125}Gd_{0.125}Nb_{0.125}O_2$ and (d) c-4 $Zr_{0.7}Y_{0.1}Gd_{0.1}Nb_{0.1}O_2$; (d) BSE images of grains of samples c-1, c-2, c-3 and c-4 after thermal etching.

significantly lower than the solid solubility limit, Y^{3+}, Gd^{3+} and Nb^{5+} could be sufficiently solid solubilized in the ZrO_2 lattice, which completely inhibited the formation of the monoclinic phase. This finding was well supported by the backscattered electron (BSE) images (see Fig. 1e) and element distribution spectrum (EDS) of thermally etched surfaces (see Fig. S1–S4, Table S5), revealing a clear correlation between rare-earth doping concentration and phase stability in ZrO_2-based ceramics. As shown in EDS analysis, significant Nb^{5+} segregation was observed at grain boundaries in high-doped c-1 and c-2, which was associated with the formation of $YNbO_4$ and matched the monoclinic phase content obtained by XRD refinement, being consistent with slow Nb^{5+} diffusion. In low-doped c-3 and c-4, uniform elemental distribution was observed, confirming a single-phase cubic structure. These results were aligned with BSE and XRD data, directly validating the presence of $YNbO_4$ and phase separation behavior. Specifically, excessive doping ($x > 0.17$) induced phase segregation, whereas a fully stabilized cubic phase could be achieved when the doping level remains below a critical threshold ($x \leq 0.1$). These results demonstrated that the rare-earth content is the decisive factor in controlling the phase composition of these materials.

3.2. Thermal properties

The thermal conductivities of $Zr_{1-3x}Y_xGd_xNb_xO_2$ across the temperature range of 673~1673 K were presented in Fig. 2a. As could be seen from Fig. 2a, the thermal conductivities of samples exhibited pronounced temperature-dependent variations, a trend strongly correlated with both the concentration of rare-earth dopants and the phase structure. Sample c-1 showed obvious thermal conductivity fluctuations due to unincorporated m-$YNbO_4$ at grain boundaries. At elevated temperatures, localized sintering of the monoclinic phase densifies grain boundaries and closes pores, reducing phonon scattering and enhancing thermal conductivity. However, the accumulation of phase transition stresses might subsequently induce microcracks, thereby intensifying phonon scattering and causing further thermal conductivity fluctuations. As the rare-earth doping concentration decreased from c-2 to c-4, the rare-earth ions were more effectively incorporated into the ZrO_2 lattice, resulting in a significant increase in the content of cubic phase. This structural evolution correlated with a progressive decline in thermal conductivity. Notably, sample c-4 demonstrated optimal thermal performance, maintaining a thermal conductivity below 1.7 W/m·K over the entire temperature range from 673 K to 1673 K and reaching a minimum of 1.37 W/m·K at 1673 K. This superior behavior was primarily attributed to the inherently low lattice thermal conductivity of the single-phase cubic structure. Interestingly, all four samples exhibited lowest thermal conductivity values at the highest temperatures, indicating promising suitability for prolonged high-temperature applications.

The phase stability of the samples was tested by holding $Zr_{1-3x}Y_xGd_xNb_xO_2$ at 1873 K for a long time, and the phase compositions of the samples after 100~300 h of heat treatment are shown in Fig. 2c, and the XRD results showed that no new phase characteristic peaks appeared in all the samples after 300 h, suggesting that the samples had excellent phase stability under extreme high temperature conditions. However, with the extension of the holding time, the diffraction peak intensities of each component showed a systematic decrease and the peak shapes were slightly broadened, which might be related to the grain coarsening and the aggravation of lattice distortion under high temperature. Despite the decrease in peak intensities, the positions and relative intensity ratios of all diffraction peaks remained unchanged, indicating that the lattice parameters and long-range order of the initial phase were not damaged.

The changes in the elastic modulus of $Zr_{1-3x}Y_xGd_xNb_xO_2$ after heat treatment at 1873 K for different times are shown in Fig. 2b. Although the elastic modulus of all the samples showed an increasing trend, the increase was significantly correlated with the doping concentration of rare earths.

The elastic modulus of the highly doped components (c-1 and c-2) had lower initial values (c-1: 108.92 GPa; c-2: 150.23 GPa), but increased dramatically to 219 GPa and 218.67 GPa with the increase of holding time, which was mainly attributed to the local densification of the unconsolidated rare-earth oxides at the grain boundaries, resulting in the enhancement of the material stiffness. In contrast, the lower-doped compositions (c-3 and c-4) exhibited higher initial elastic moduli (c-3: 192.07 GPa; c-4: 199.78 GPa), but demonstrated only

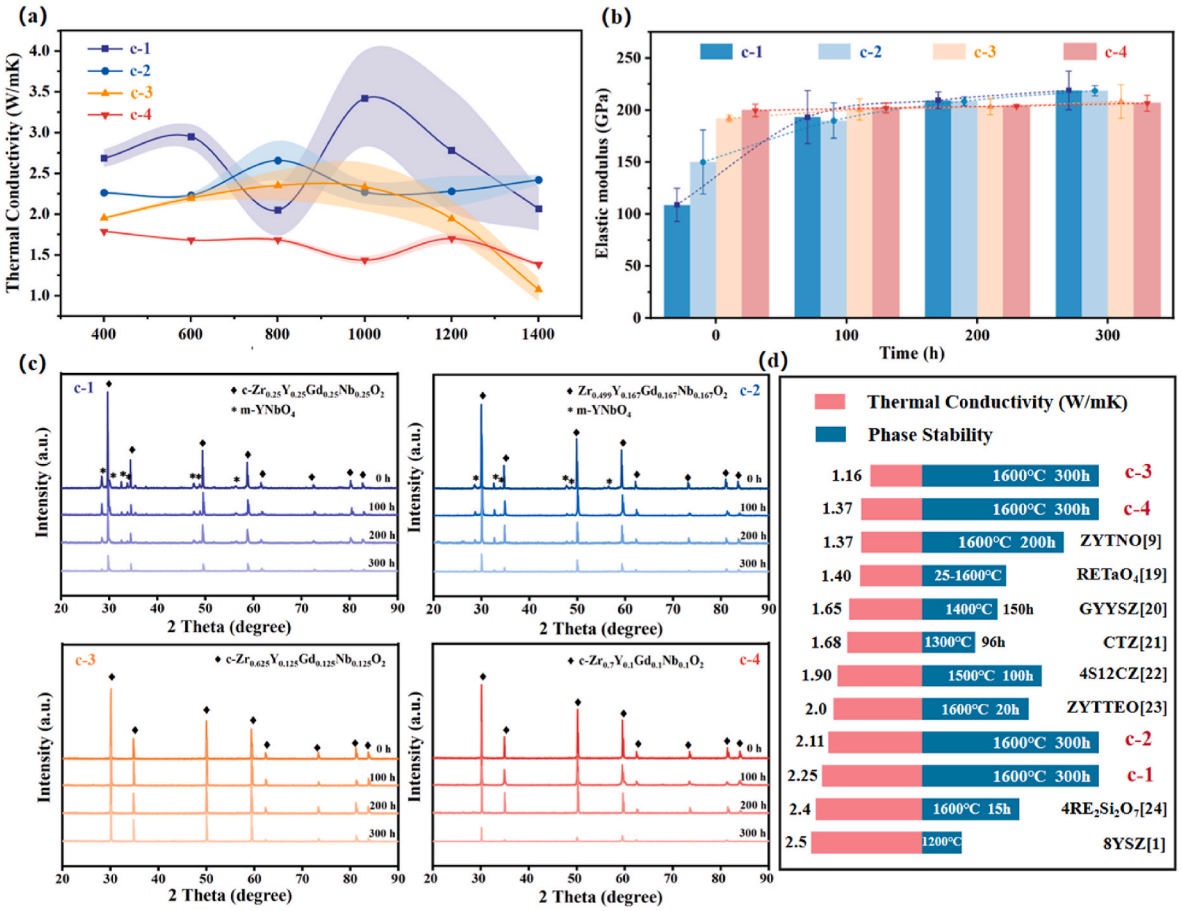


Fig. 2. Thermal properties: (a) Thermal conductivity of $Zr_{1-3x}Y_xGd_xNb_xO_2$; (b) Elastic modulus of $Zr_{1-3x}Y_xGd_xNb_xO_2$ after heat-treated at 1873 K with different times; (c) XRD patterns of $Zr_{1-3x}Y_xGd_xNb_xO_2$ after heat-treated at 1873 K with different times; (d) Phase stability and thermal conductivity of $Zr_{1-3x}Y_xGd_xNb_xO_2$, compared with current material systems: $Zr_{1-4x}Y_xYb_xTa_xNb_xO_2$ (ZYTNO), $RETaO_4$ (RE = Nd, Eu, Gd, Dy, Er, Yb, Lu), Gd-Yb-3.5YSZ(GYYSZ), ZrO_2-CeO_2-TiO_2 (CTZ), $Sc_{0.04}Ce_{0.12}Zr_{0.84}O_{1.98}$ (4S12CZ), $Zr_{0.544}Y_{0.114}Ta_{0.114}Er_{0.114}Ti_{0.114}O$ (ZYTTEO), $4RE_2SiO_7$ (RE = Y, Yb, Er, Sc).

modest increases (10~20 GPa after 300 h). This behavior suggested that the complete solid solution of Y^{3+}, Gd^{3+}, and Nb^{5+} effectively suppressed grain-boundary segregation and pore closure, thereby mitigating stress concentration induced by grain growth and consequently slowing the elastic modulus enhancement rate. In addition, as shown in Fig. 2d, compared to conventional YSZ [1] and existing high-performance TBC candidates, such as $Zr_{1-4x}Y_xYb_xTa_xNb_xO_2$ (ZYTNO) [9], $RETaO_4$ (RE = Nd, Eu, Gd, Dy, Er, Yb, Lu) [19], Gd-Yb-3.5YSZ(GYYSZ) [20], ZrO_2-CeO_2-TiO_2 (CTZ) [21], $Sc_{0.04}Ce_{0.12}Zr_{0.84}O_{1.98}$ (4S12CZ) [22], $Zr_{0.544}Y_{0.114}Ta_{0.114}Er_{0.114}Ti_{0.114}O$ (ZYTTEO) [23], $4RE_2SiO_7$ (RE = Y, Yb, Er, Sc) [24], $Zr_{1-3x}Y_xGd_xNb_xO_2$ (especially the low-doped components c-3 and

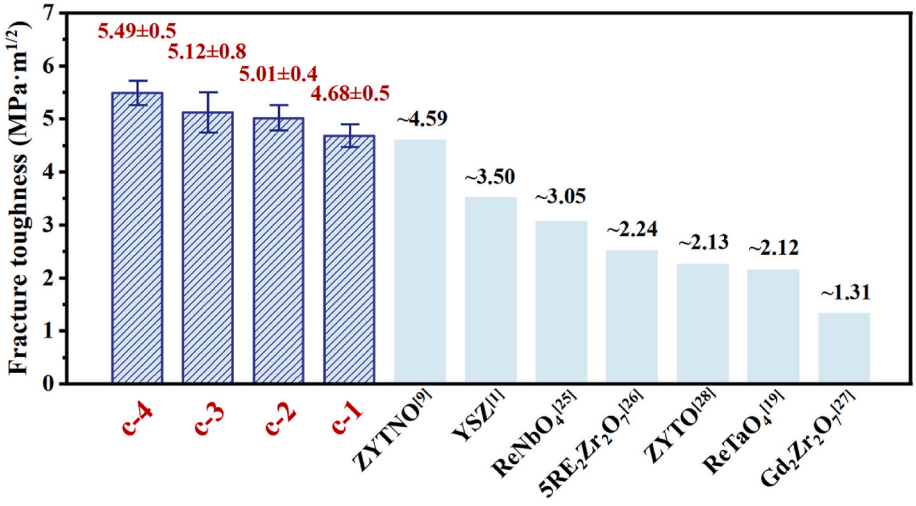


Fig. 3. Fracture toughness of $Zr_{1-3x}Y_xGd_xNb_xO_2$, compared to most of current material systems: $Zr_{1-4x}Y_xYb_xTa_xNb_xO_2$ (ZYTNO), YSZ, $RENbO_4$ (RE = Nd, Sm, Gd, Dy, Er, Yb), $5RE_2ZrO_7$ (RE = Sm, Eu, Tb, Dy, Lu), ZrO_2-$YO_{1.5}$-$TaO_{2.5}$ (ZYTO), $RETaO_4$ (RE = Nd, Eu, Gd, Dy, Er, Yb, Lu), $Gd_2Zr_2O_7$.

c-4) exhibited remarkable low thermal conductivity with excellent high-temperature phase stability. Its high-temperature thermal conductivity (<1.4 W/m·K) was approximately 44 % lower than that of YSZ (~2.5 W/m·K) [1], and the material retained a stable single-phase cubic structure even after prolonged thermal exposure at 1873 K for 300 h.

3.3. Fracture toughness

As shown in Fig. 3, the fracture toughness of $Zr_{1-3x}Y_xGd_xNb_xO_2$ was compared with various state-of-the-art TBC materials[25–28]. Notably, c-4 exhibited the highest fracture toughness (5.49 MPa·$m^{1/2}$) among all the materials evaluated, significantly outperforming most newly developed TBC materials which typically demonstrated inferior fracture toughness compared to conventional YSZ.

The fracture toughness of $Zr_{1-3x}Y_xGd_xNb_xO_2$ ceramics, as determined by Vickers indentation measurements, exhibited a strong dependence on rare-earth doping concentration. Fracture toughness demonstrated a progressive increment from 4.68 to 5.49 MPa·$m^{1/2}$ with decreasing rare-earth doping concentration (c-1 to c-4). This enhancement in fracture toughness was closely correlated with the evolution of phase composition and microstructure. Particularly, the superior fracture toughness was observed in the fully cubic-phase stabilized compositions (c-3 and c-4, >99 % cubic phase) compared to their biphasic counterparts (c-1 and c-2).

As shown in Fig. 1a~e, in highly doped compositions (c-1 and c-2), the presence of monoclinic phase segregation at grain boundaries, as identified through XRD and SEM analysis, induced the formation of fine-grained regions. These microstructural features acted as stress concentration sites, thereby reducing the energy dissipation efficiency during fracture and resulting in lower toughness values.

In contrast, the low-doped compositions (c-3 and c-4) benefited from their single-phase cubic structure and homogeneous microstructure. The retained single-phase cubic structure ensured homogeneous stress distribution, free from secondary-phase pinning effects. Furthermore, the homogeneous solid solution of rare-earth elements enhanced the grain boundary strength of the cubic phase, resulting in high grain boundary bonding that effectively resisted crack propagation and contributed to the superior fracture toughness [29].

3.4. CMAS corrosion resistance

The CMAS corrosion tests were carried out at 1673 K for 4 h. The cross-sectional SEM images of the corroded samples are shown in Fig. 4a. It could be seen from Fig. 4a that the uniform CMAS infiltration layers were formed on the surfaces of samples, and the infiltration layer of the highly doped component c-1 (x = 0.25) was shallower (65.4 μm). The surface generated a rod-like apatite phase ($Ca_2RE_8(SiO_4)_6O_2$), but its structure was loose and unevenly distributed. This phenomenon was closely related to the segregation of excess rare earth elements (Y^{3+}, Gd^{3+}, and Nb^{5+}) at the grain boundaries in c-1. A large number of unincorporated rare earth ions at grain boundaries preferentially reacted with CMAS to form apatite phases, leading to the inability of apatite phases to continuously cover the surface.

As the rare earth content decreased (c-2 to c-4), the material gradually transitioned to a single cubic phase dominated structure. Furthermore, the reduction of unincorporated rare earth oxides at grain boundaries inhibited apatite phase formation. The diffusion of CMAS melt along the grain boundaries directly reacted with the substrate, which triggered the dissolution of the grains and the formation of spherical particles. Due to the lack of reactive protective layer, the depth of penetration of CMAS increased significantly (reaching up to 180.3 μm in c-4). When the test time was extended to 50 h at 1673 K, the resulting corrosion curves are shown in Fig. 4b. The penetration behavior of the four components of $Zr_{1-3x}Y_xGd_xNb_xO_2$ could be divided into an infiltration-dominated stage and a reaction-limited stage [30,31]. In the initial stage of infiltration, CMAS rapidly wet the pores and grain boundaries on the material surface due to its low viscosity and high fluidity. At this stage, the infiltrated layer was thin and porous, failing to effectively hinder melt diffusion, thereby leading to a high infiltration rate (e.g., 18.9 μm/h in the early phase of c-4). As time progressed, the reaction layer gradually thickened and densified, accompanied by grain coarsening and pore reduction. This microstructural evolution narrowed or even blocked internal diffusion pathways, thereby increasing resistance to CMAS infiltration. Concurrently, the viscosity of the CMAS melt rose as the reaction proceeded, deteriorating its fluidity and leading to a significant decline in the infiltration rate (<1 μm/h). The time required for different compositions to transition into the reaction-limited stage varied due to differences in their phase structure and chemical

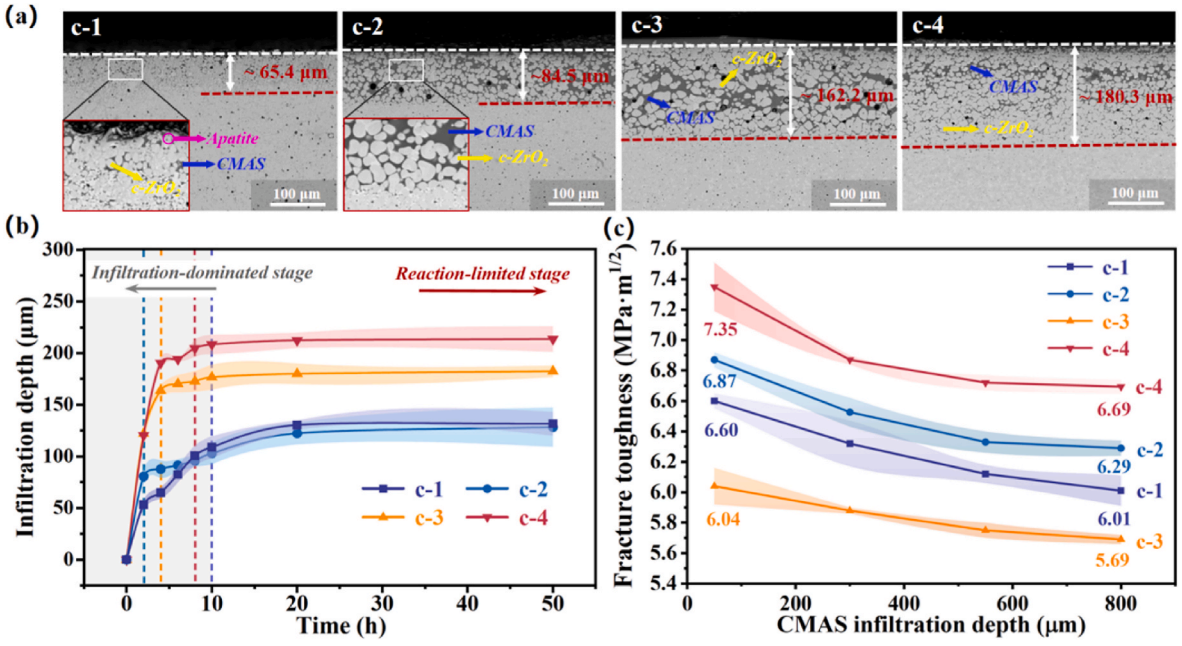


Fig. 4. CMAS corrosion resistance: (a) Cross-sectional SEM images of samples after corrosion at 1673 K for 4 h; (b) CMAS corrosion curves for 0~50 h at 1673 K; (c) Fracture toughness of samples at different depths after CMAS corrosion.

reactivity.

After CMAS corrosion at 1673 K for 4 h, the fracture toughness of all four $Zr_{1-3x}Y_xGd_xNb_xO_2$ compositions exhibited an increasing trend, as shown in Fig. 4c. Notably, while the CMAS-enriched reaction layer exhibited extremely low fracture toughness, the region approximately 50 μm beneath it demonstrated a significant toughness enhancement compared to the uncorroded state (e.g., c-4 increased from 5.49 $MPa{\cdot}m^{1/2}$ to 7.35 $MPa{\cdot}m^{1/2}$). This improvement could be attributed to microstructural optimization and stress redistribution induced by CMAS corrosion. With increasing depth (300~800 μm), where the corrosion effect diminished, the fracture toughness gradually decreased but remained higher than that of the original substrate. These observations indicated that the detrimental effects of CMAS corrosion on mechanical properties were primarily localized within the surface reaction layer. In contrast, the underlying matrix region, which experienced limited CMAS infiltration, exhibited improved mechanical properties due to optimized stress distribution and enhanced grain boundary bonding.

4. Conclusions

A novel series of $Zr_{1-3x}Y_xGd_xNb_xO_2$ medium-entropy oxide ceramics was prepared. The corresponding thermal-mechanical properties and CMAS corrosion resistance were systematically studied. The main conclusions are as follows:

(1) Low-doping compositions (x = 0.1, 0.125) maintain a stable single cubic phase through effective rare-earth solid solution, achieving low thermal conductivity of 1.07~1.37 W/m·K at 1673 K, approximately 44 % lower than that of conventional YSZ.
(2) All four compositions demonstrate exceptional high-temperature phase stability, maintaining their structural integrity without phase transformation after 300 h at 1873 K.
(3) The materials exhibit significantly improved mechanical properties, including a high fracture toughness of 5.49 $MPa{\cdot}m^{1/2}$ and excellent elastic modulus stability during prolonged high-temperature exposure (<0.07 GPa/h increase rate).
(4) The materials exhibit exceptional CMAS corrosion resistance at 1673 K, demonstrating remarkably low penetration rates (<1 μm/h). Remarkably, the fracture toughness after corrosion increases to 7.35 $MPa{\cdot}m^{1/2}$, suggesting in-service performance improvement under extreme conditions.

CRediT authorship contribution statement

M.T. Lin: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **J.H. Zu:** Investigation, Formal analysis. **Y. Wang:** Writing – review & editing. **L. Zhang:** Methodology, Data curation. **F.L. Yu:** Writing – review & editing. **M. Liu:** Supervision. **H.D. Wang:** Validation, Funding acquisition. **Y. Bai:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ceramint.2025.09.008.

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